

Data Science vs Machine Learning vs AI

Agenda

- ① Define
 - ② Core components & methodologies
 - ③ Applications & Use-cases
 - ④ Tools & Tech
 - ⑤ Projects to build your portfolio.
- + links for practicing

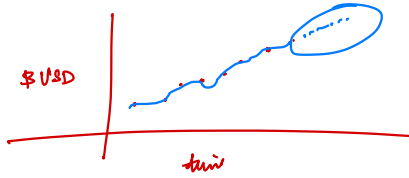
Raw Data → websites, sensor, transactional, Aadhar

Data Science

↳ exploring, cleaning & interpreting the data

Machine Learning

↳ Building, training
evaluation



AI

Subject that includes anything that allows machines to mimic human intelligence.

→ Is machine learning part of AI?

Applications

① Data Science → Business Analytics
Revenue, operation margin, profit.
Daily Signups, DAU, MAU

→ Customer Segmentation

→ Risk Assessment.

→ Anomaly detection

② Machine Learning

- Recommendation Systems.
- Predictive Maintenance
- Forecasting
- Fraud detection

③ AI

Self driving cars, Chatbots, language translation.
Medical diagnosis.

Core Components

Data Science

 → Data extraction

→ Data cleaning / Data wrangling

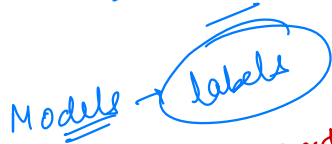
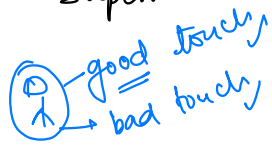
→ Data analysis →

→ Data Visualization, Dashboard

→ Domain expertise

Machine Learning

Supervised



Eg Regression → predict number
Classification → predict group

eg Mail → spam or ham

Unsupervised

Discover
hidden
patterns

→ unlabelled data

eg clustering
grouping similar
things together.

Reinforcement
Learning

Train agents in a
dynamic environment.

Robotics
→ gaming

AI



① Natural Language Processing (NLP)

- + human language (analyse)
- + power chatbots, translators } generative AI

② Computer Vision (CV)

- + Face screen unlocking
- + VR
- + AR
- + self driving cars
- + image recognition.

Tools & Tech / Skills

✓ Data Science

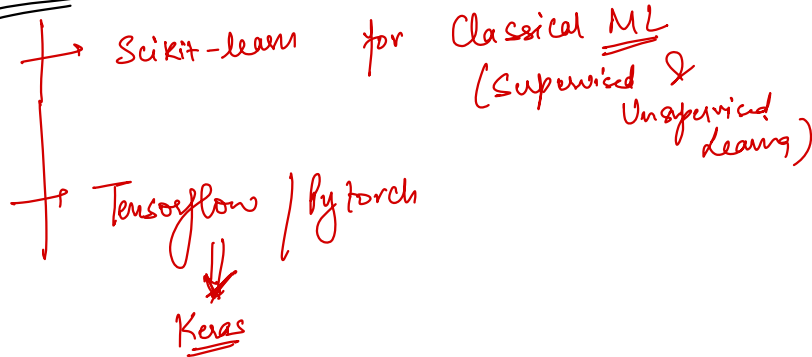
- SOL (Structured Query Language)
- Python (libraries → Pandas, Numpy, Matplotlib, Seaborn)

Visualization → Tableau / Power BI

Machine Learning

(Pre-requisite → Data Science)

Frameworks



AI

- + NLP → Huggingface Transformer, Spacy
- + CV → OpenCV, YOLO
- + Advanced AI → Langchain, LlamaIndex
CrewAI

Roles

Data Scientist

Skills → Data analysis, SQL, Storytelling

Consulting Companies → BCG, Mck, Bain

Data Scientist, Business Analyst, Data Analyst,

Machine Learning

+ Strong programming skills

→ Model deployment (Cloud Technologies,
AWS/GCP/Azure)

Roles

ML engineer, Research Scientist

AZ

→ AI engineer, deep learning specialist, robotics engineer

Projects → Kaggle.com

Learning → Scaler-topic

Certificates → HackerRank.com

Resume →

<https://www.overleaf.com/gallery/tagged/cv>